

Charlie Jacobs

Cleveland, OH • (202)-748-4368 • cjj72@case.edu

Junior at Case Western Reserve University pursuing a B.S. in Biochemistry with a concentration in Computational Health Sciences, interested in both computational and wet-lab roles in pharmaceutical research and development. Currently completing graduate-level coursework in preparation for applying to the Combined Master's in Biotechnology program with a concentration in Pharmacology, with an interest in interdisciplinary research spanning computational biology, molecular biology, and drug development.

Skills

Bioinformatics: R Programming • PPI Network Analysis • Network Alignment • Association Rule Mining • Gene Ontology Analysis • Reactome Pathway Analysis • Large-Scale Data Manipulation

Molecular Biology: Bacterial Transformation & Culture • Plasmid Purification • PCR • Gel Electrophoresis

Documentation & Quality: cGMP Principles • Good Documentation Practices (GDP) • Data Integrity (ALCOA+) • SOP Adherence • Electronic Laboratory Notebooks (LabArchives) • Experimental Reproducibility • Scientific Writing

Research & Laboratory Experience

Undergraduate Research Assistant

Bebek Lab, Dep. Nutrition CWRU School of Medicine

Cleveland, OH

April 2026 - Present

- First-authored a peer-reviewed paper accepted for publication and presentation at the Pacific Symposium on Biocomputing (PSB) 2027; designed and implemented the computational pipeline used to formulate colorectal cancer signaling networks, including construction of a coexpression-filtered human protein-protein interaction network and Gene Ontology-based association rule mining.
- Currently working on an R package that computes semantic similarity for Reactome pathway data to aid in functional pathway analysis.

Biotechnology Laboratory

Case Western Reserve University School of Medicine

Cleveland, OH

August - December 2026

- Applied bacterial transformation and culture, plasmid construction and purification, PCR, restriction digestion, ligation, and agarose gel electrophoresis in a graduate-level molecular biology laboratory course.
- Conducted DNA quantification and sequence analysis while maintaining detailed records in laboratory and electronic notebooks to support reproducibility.

Work & Leadership Experience

Swim Coach

Montgomery County Swim League (MCSL)

Montgomery County, MD

Summers 2022 - 2026

- Coached swimmers ages 4-18 across three MCSL teams: Rock Creek Pool, Quail Valley Pool, and Garrett Park Pool.
- Planned and ran daily practices and prepared swimmers for weekly A/B meets and championship competitions.
- Took on increasing responsibility for practice planning, meet organization, parent communication, and supervision of junior coaching staff.

Co-President

CWRU Club Swim

Cleveland, OH

January 2026 - Present

- Co-lead a 12-member Executive Board and 6-member subcommittee, coordinating weekly meetings, swim meets, team activities, and charity events.
- Led practices for 50+ swimmers and fostered a welcoming, competitive environment for athletes of all experience levels as Team Captain ('25-'26)

Education

Case Western Reserve University

Biochemistry B.S. with Concentration in Computational Health Sciences

Relevant coursework: Organic Chemistry, Pharmacokinetics, Pharmacodynamics,

Statistics, Development & Physiology

Cleveland, OH

Expected 2028